

Project Milestone 4 – Technical Brief Draft

To: Avery D. Lion, President, Natural Catalysts Inc.

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RE: Technical Brief of Final Result

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Introduction

The client, Natural Catalysts Inc. asks for help characterizing the functioning of their enzymes and also in responsibly marketing them to the market. According to the team, the client needs an algorithm to model the values of the enzymes and generate V_0 , $V_{\max}$ and K_m values. This includes graphs, code and supporting error analyses. The primary constraint was the timeline provided, and other challenges include working exclusively using the enzyme models provided, without coming up with non-linear regression models.

The algorithm we built works through these following steps:

Using the initial data, the algorithm first finds the V_0 values. We have picked the Lineweaver-Burk method to find our $V_{\max}$ and K_m values, so we first find the appropriate x and y variable values for the same. Using these values, we find the linear plot using polyfit and polyval functions. As we know the intercept and slope, we calculate the $V_{\max}$ and K_m values from these values.

Point 1: Usage of Lineweaver Burk

To find the $V_{\max}$ and K_m values, we had to pick a particular model which we would use for it, like the Michaelis-Menten, Eadie-Hofstee, Lineweaver-Burk. When first picking what model we wanted to use, we did extensive research on what model seemed to be used the most amongst researchers studying enzymes. We also checked which set of V_0 plots seemed the easiest to linearize, and then picked the Lineweaver - Burk method for our algorithm.

Point 2: Automated method to determine linear region

First, we decided that we would calculate the slopes at each data point and find the percent difference between the current point and the previous point. We would stop the linear region when the percent difference went over a certain threshold. This was a critical decision because it made our process of determining the linear region automated and also made it versatile.

Point 3: Determination in linear region using percent difference in slopes

To overcome this issue of noise, instead of determining the slopes at each point, we decided to calculate the slopes for a set of points. Thus, we would get the slope between the first and the last point of that region which would be a good estimate of the average slopes of those points. This way, with a large enough region, the issue of noise would completely be overcome.

Parameter Identification Procedure

First, to calculate the v_0 values we created an algorithm which first finds the points which lie in the linear region of the concentration v/s time graph and then finds the average slope of the linear region. Initially, we divided our dataset into regions of 10 points which reduced the noise. We then created a loop which iterates through the dataset taking 10 points at a time which could calculate the graph. Then

the percent difference in the slope of the current region and the first region was calculated. If this percent difference was greater than a threshold, (determined to be 20%), then the loop would end. Finally, we determined v_0 by calculating the slope between the first and last point of the linear region.

We have used the Lineweaver Burk method to determine the values of K_m and V_{max} . This method involves plotting the reciprocal of the reaction velocity against the reciprocal of the substrate concentration. We then used the polyfit function to calculate the slope and intercept of the linear equation. The value of V_{max} is the reciprocal of the intercept, and the value of K_m is the slope times V_{max} . Using this, we can calculate K_m and V_{max} .

Results

Enzyme	A	A_dup	B	B_dup	C	C_dup	D	D_dup	E	E_dup
V_{max} ($\mu\text{M/s}$)	0.8794	0.8775	0.6605	0.4497	1.4481	1.5572	1.6109	3.1158	1.8603	1.2688
K_m (μM)	140.6586	136.421	249.3627	159.9406	222.1799	234.0239	289.6303	568.9891	179.9794	113.9194

Table 1: The V_{max} and K_m values for all the enzymes and their duplicates

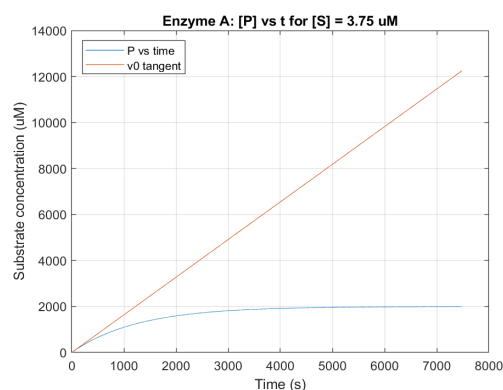


Figure 1: Sample Substrate concentration versus time graph for Enzyme A data

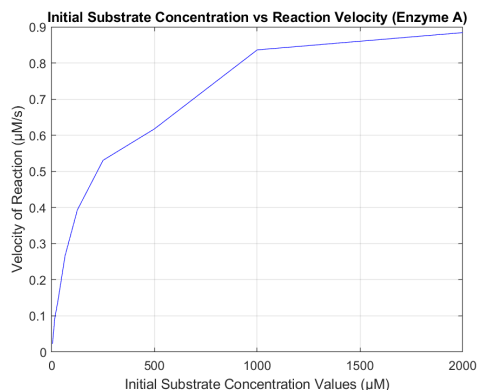


Figure 2: Velocity of reaction versus initial substrate concentration for Enzyme A data

Interpretation

Although our algorithm is extremely accurate and predicts values within a few percent of the actual values, the error that occurs in our program mainly lies with the usage of Lineweaver Burk equation. This method relies on linearizing the data, which may result in some calculations giving slightly different values. The experiment was not conducted with great precision or accuracy, since the values of the duplicate tests and the original tests have high percent errors, indicating that the tests were not conducted using the exact same conditions. The parameter identification algorithm however, produces highly accurate results. Table 3 of the Answer Sheet displays the percent error between the values generated by the algorithm and the actual values. It shows that the percent error values are extremely low, indicating that the results are highly accurate.

Based on the quality of experiments and the accuracy of the parameter identification algorithm, NaturalCatalysts can claim that their enzymes have been rigorously tested and their enzymes have shown to be effective in breaking down substrates that produce stains on fabric. NaturalCatalysts can claim that their enzymes are produced to the highest standards of quality and reliability to ensure consistent performance for their customers.

References

Libretexts. (2021, March 6). 4.10: Lineweaver-Burk Plots. Biology LibreTexts. Retrieved April 28, 2023, from [https://bio.libretexts.org/Bookshelves/Biochemistry/Book%3A_Biochemistry_Free_and_Easy_\(Ahern_and_Rajagopal\)/04%3A_Catalysis/4.10%3A_Lineweaver-Burk_Plots](https://bio.libretexts.org/Bookshelves/Biochemistry/Book%3A_Biochemistry_Free_and_Easy_(Ahern_and_Rajagopal)/04%3A_Catalysis/4.10%3A_Lineweaver-Burk_Plots)

polyfit command. Polynomial curve fitting - MATLAB. (n.d.). Retrieved April 28, 2023, from <https://www.mathworks.com/help/matlab/ref/polyfit.html>

Appendix: Figures and Tables

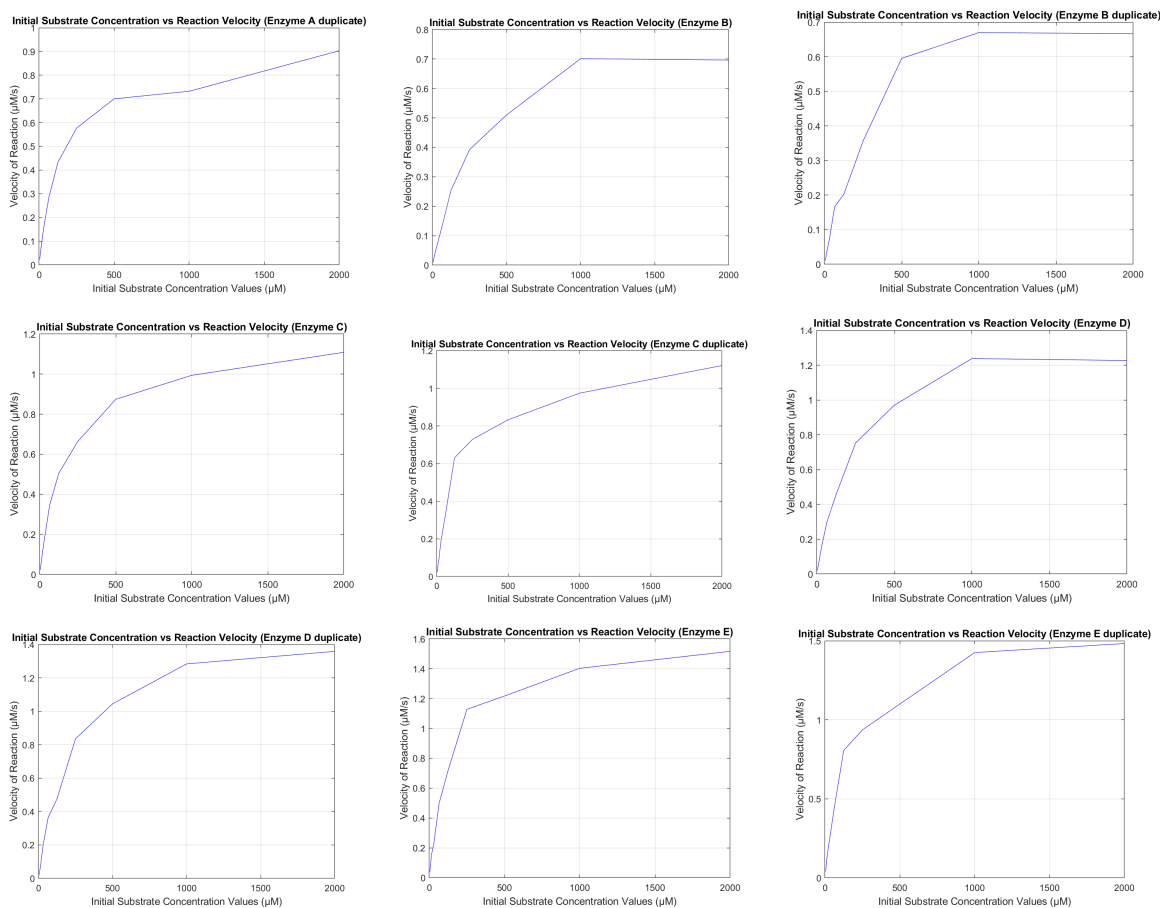


Figure 3-11: Velocity of reaction versus initial substrate concentration for Enzyme A duplicate, Enzyme B, Enzyme B duplicate, Enzyme C, Enzyme C duplicate, Enzyme D, Enzyme D duplicate, Enzyme E, Enzyme E duplicate respectively